

Harsh Amin

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EDUCATION

University of British Columbia

BSc Co-op Computer Science · Math

GPA: 4.00

April 2027

Vancouver, BC

Awards: Trek Academic Excellence Scholarship - awarded to top 5% of faculty

Extracurricular: UBC Launch Pad Design Team, Intramural Hockey, Weight Lifting Club, Entrepreneurship Club

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Racket, R, HTML/CSS, JavaScript, TypeScript, SQL, GraphQL

Frameworks: LangChain, LlamaIndex, Agents, Ollama, React, Flask, FastAPI, Node.js/Deno, Express.js, SQLAlchemy

Developer Tools: AWS, AWS CDK, REST, OpenAPI, PostgreSQL, MongoDB, Docker, Git, Linux/Unix, Jenkins

EXPERIENCE

ML Software Development Engineer Intern

Amazon - AI Center of Excellence

May 2026 – August 2026

Vancouver, BC

- Reduced agentic simulation scenario authoring time by **85%** by building a test-generation platform, packaged as an **AWS CDK** construct that uses a production agent's live traffic into approval-gated regression tests.
- Eliminated **95%** of redundant test cases by building a trace-clustering pipeline using **BGE-M3** embeddings and an ensemble density-based clustering approach, distilling **8,000+** production agent executions into **380** distinct behavioural scenarios across **2** structurally different agents (a RAG assistant and a sequential tool-use agent).
- Cut the regression-escape rate by **40%** by designing a behavioural-drift detector that auto-generates new simulations when novel production behaviour emerges, gated by a review dashboard on **CloudFront** and **Fargate**
- Improved behavioral-cluster coherence by **20%+** by introducing a sequence-aware distance metric (**Dynamic Time Warping**) that groups agent runs by execution order rather than tool set alone.

GenAI Software Developer Intern

UBC Cloud Innovation Centre (Collab. w/Amazon Web Services)

May 2025 – Dec 2025

Vancouver, BC

- Built **open source Generative AI** applications **end-to-end** for **3+** sponsor teams, leveraging **AWS Bedrock Guardrails, IAM, WAF, Shield, Cognito, Secrets Manager** to deliver secure and scalable **AI/ML** solutions
- Designed and deployed an enterprise-grade **AI Agent** using an MCP server; integrated **DynamoDB** for continuous conversational memory, sustaining context across deep multi-turn sessions and improving query precision by **30%**.
- Migrated from unreliable local Docker workflows to a standardized **AWS CI/CD** pipeline (**CodeBuild, CodePipeline, ECR**), cutting cross-environment inconsistencies by **33%** and reducing deployment time by **40%**
- Collaborated with a cross-functional team of **5+** project developers, managers, and stakeholders, presenting technical solutions and documenting each phase for seamless deployment and knowledge transfer

Co-president and Ex. Technical Software Team Lead

UBC Launchpad - Project: Forum AI (LLM based student question board for UBC classes)

Sep 2024 – Present

Vancouver, BC

- Designed the **Retrieval Augmented Generation** model and updated it to be **20%** quicker
- Co-lead a **15+** member team of developers and designers, arranging weekly syncs and tickets over 8 months
- Migrated to **FastAPI** endpoints and implemented trigger-based embedding retrieval from a **PGVector** database, resulting in a **27%** reduction in API response time
- Led the migration to the **Deno** runtime, optimizing containerized function performance and achieving a **20%+** improvement in execution speed within **Docker** environments for resource heavy functions (ex. RAG)

TECHNICAL PROJECTS

OpenEd - AI Study Companion | AWS CDK, WAF, RDS, Bedrock, LangChain, Docker, RAG, OpenAPI Dec 2025

- Developed a cloud-native application sponsored by BCcampus that will enable **100,000+** students/teachers to streamline studying, teaching, and learning material formation by **~ 30%**
- Built a textbook-specific chat assistant leveraging **AWS Bedrock** and a **LangChain RAG** pipeline, scaled to process **750+** textbooks with **15+** hours of audio/video transcripts, storing embeddings in **RDS PostgreSQL**
- Deployed a production-ready **API Gateway** stack from an **OpenAPI** spec, integrating **10+** **Lambda** functions (via **Docker** images and **Lambda Layers**) with automated **CI/CD** integration scaling to **1K+** requests/day